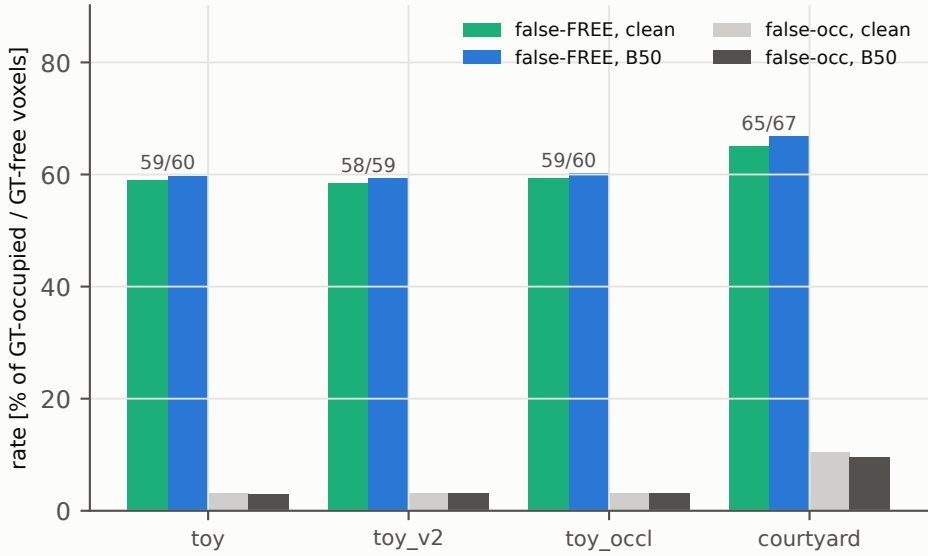


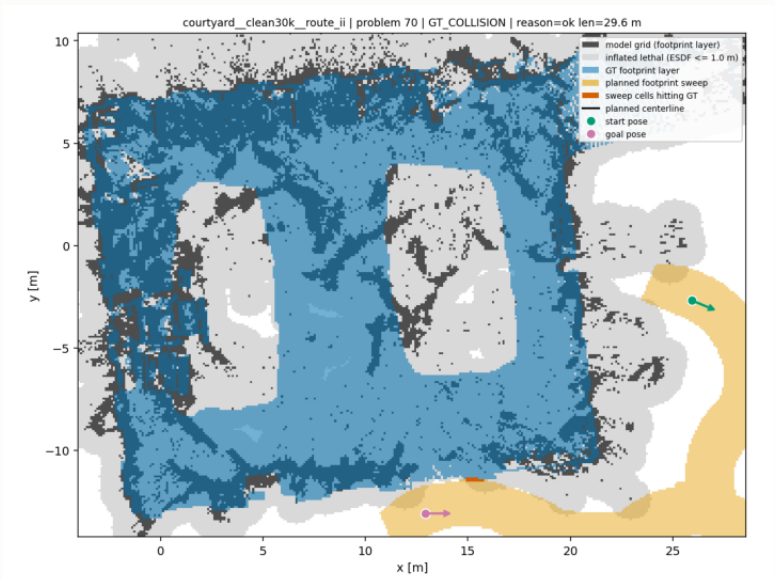
(a) d1 occupancy confusion @0.10 m — B50 PRESERVES the clean model's (bad) confusion; false-free is safety-critical



(b) N=500 paired planner problems (seed 0; GOAL#015): clean↔B50 outcome sets IDENTICAL (CIs [0,0]); raw grids are almost-always infeasible AND unsafe when they do plan

cell	found	coll./100 plans	spurious infeas.
toy GT ref	500/500	1.8	0.0%
toy clean rt-i	58/500	0.0	88.4%
toy B50 rt-i	58/500	0.0	88.4%
cyard GT ref	494/500	1.6	1.2%
cyard clean rt-i	2/500	100.0	99.6%
cyard B50 rt-i	2/500	100.0	99.6%
cyard clean rt-ii	118/500	5.9	76.4%
cyard B50 rt-ii	118/500	5.9	76.4%

(c) maneuver example — courtyard route-ii plan sweeps a GT wall its grid marks free (R3.c panel; grid bit-identical clean↔B50)



(d) R3.b certification sheds load-bearing surface — courtyard footprint before/after (kept set EXACTLY identical clean↔B50)

